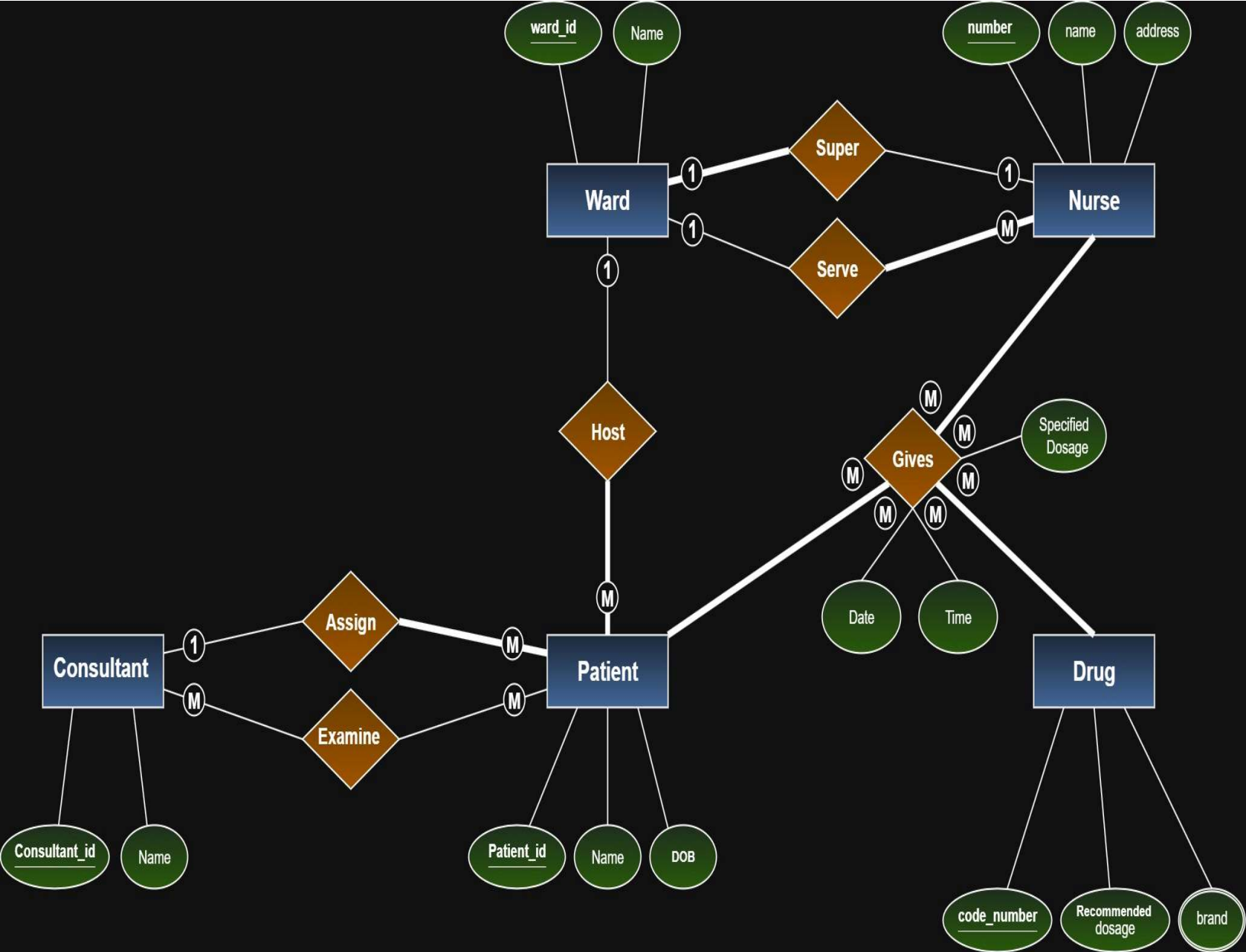


TASK:

- A General Hospital consists of a number of specialized wards. Each ward is described by ward_id, Name.
- The system records the following details about patients: Patient_id, name, Date_Of_Birth.
- Each ward may host more patients and each patient is hosted by only one ward.
- Each patient is assigned to one leading consultant but may be examined by other consultants, if required.
- Each consultant may be assigned zero or more patients and may examine zero or more patients.
- Consultants are described by Consultant_id, Name.
- The system has to record all required data each time the Nurse gives a patient a certain drug with specified dosage at certain date and time.
- Each ward is under supervision of one nurse and a nurse may supervise only one ward.
- Each Nurse must serve in one ward and ward can have many nurses.
- Data about the nurse is recorded as her name and her number and her address.
- A drug has code number, recommended dosage and more than one brand name.



Mapping in ERD

1. Mapping of Regular / Strong Entity
2. Mapping of Weak Entity
3. Mapping of Binary / Unary 1:M Relationship
4. Mapping of Binary / Unary M:M Relationship
5. Mapping of Binary / Unary 1:1 Relationship
6. Mapping of Ternary Relationship

Mapping in ERD

1. Mapping of **Regular / Strong** Entity

- **Simple Attribute:** Map it directly as a column.
 - **Derived Attribute:** Do not map it.
 - **Composite Attribute:** Decompose it into its components and map them separately.
 - **Multi-valued Attribute:** Create a new table containing the entity's PK as a FK.
The new table has a Composite PK consisting of the entity's PK and the multi-valued attribute.
-

2. Mapping of **Weak** Entity

- Add the PK of the Strong Entity to the Weak Entity as a FK.
 - The PK of the Weak Entity is a Composite PK, consisting of:
 - The PK of the Strong Entity.
 - The Partial Key of the Weak Entity.
-

3. Mapping of **Binary / Unary 1:M** Relationship

- Add the PK of the 1-side to the M-side as a FK.
- Add the Relationship Attributes to the M-side table.

4. Mapping of **Binary / Unary M:M** Relationship

- Create a new table for the relationship.
 - Add the PKs of both entities as FKs.
 - The PK of the new table is a Composite PK consisting of both PKs.
 - Add the Relationship Attributes to the new table.
-

5. Mapping of **Binary / Unary 1:1** Relationship

- May – Must: Add the PK of the May side to the Must side as a FK.
 - May – May: Add the PK of either side to the other side as a FK.
 - Must – Must: Merge the two tables into one table.
 - Add the Relationship Attributes to the table containing the FK.
-

6. Mapping of **Ternary** Relationship

- Create a new table for the ternary relationship.
- Add the PK of each participating entity to the new table as FKs.
- The PK of the new table is determined by the cardinality constraints of the ternary relationship.
- Add the Relationship Attributes to the new table.

Mapping :

Important Note :

■ Highlighted Text → PK

● Underlined Text → FK

Ward (**ward_id** , name , Super nurse number)

patient (**patient_id** , name , DOB , ward_id , consultant_id)

Consultant (**consultant_id** , name)

Examine (**patient_id** , **consultant_id**)

Nurse (**nurse_number** , name , address , ward_id)

Drug (**code_number** , Recommended dosage)

Drug_Brand (**code_number** , **brand**)

Gives_Drug (**nurse_number** , **patient_id** , **code_number** , date , **time** , Specified Dosage)